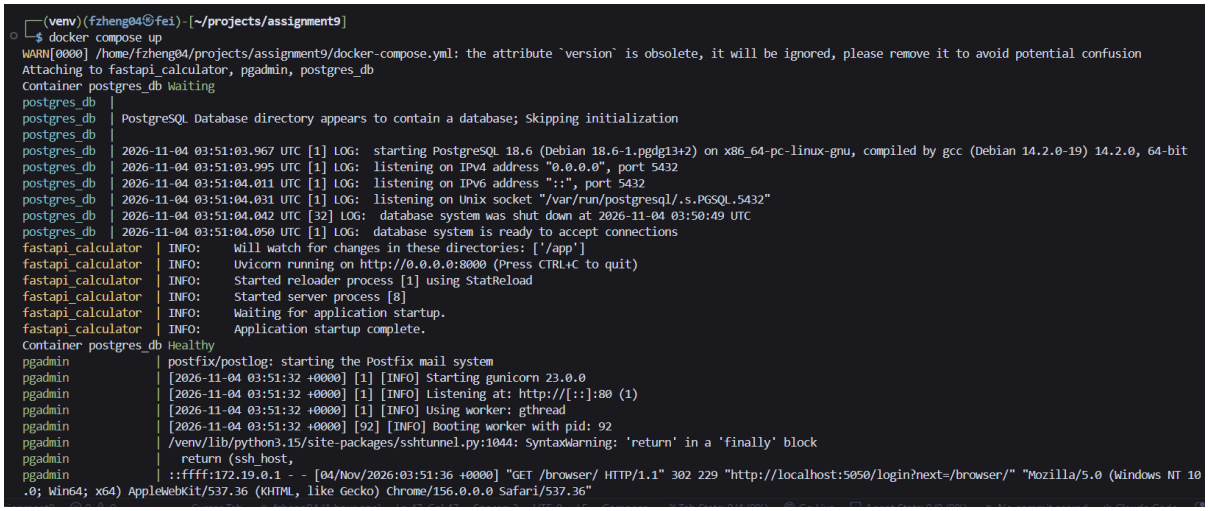
Module 9: Working with Raw SQL in pgAdmin

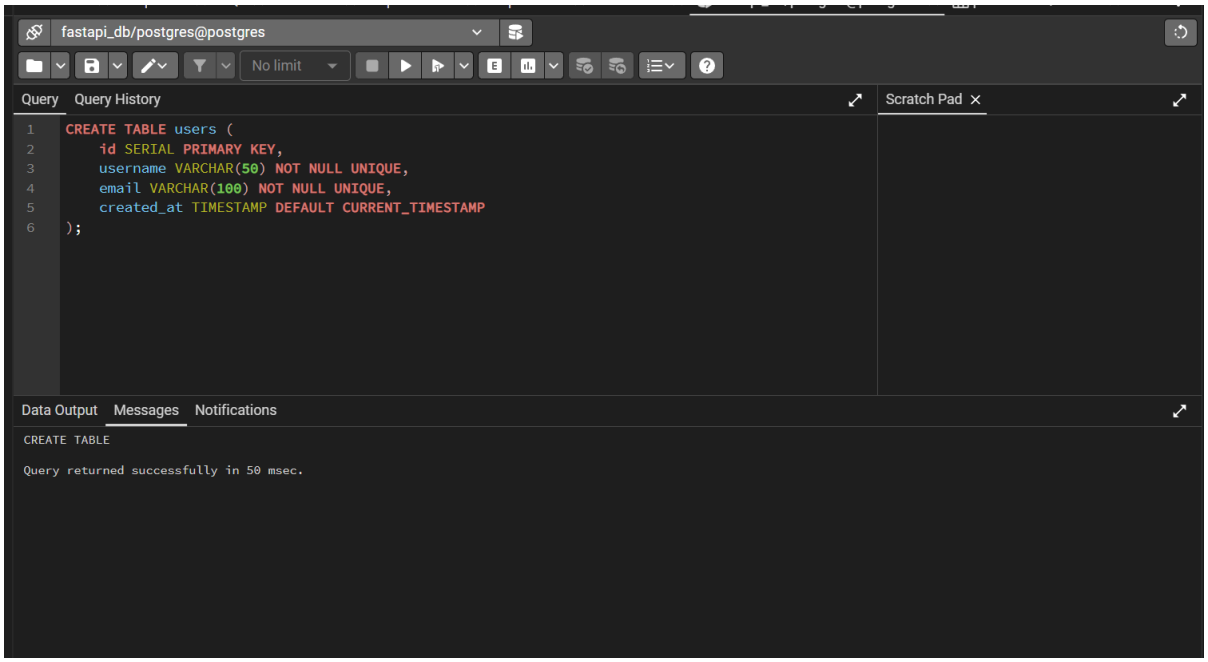
GitHub repository: https://github.com/fzheng04/assignment9

Environment Setup

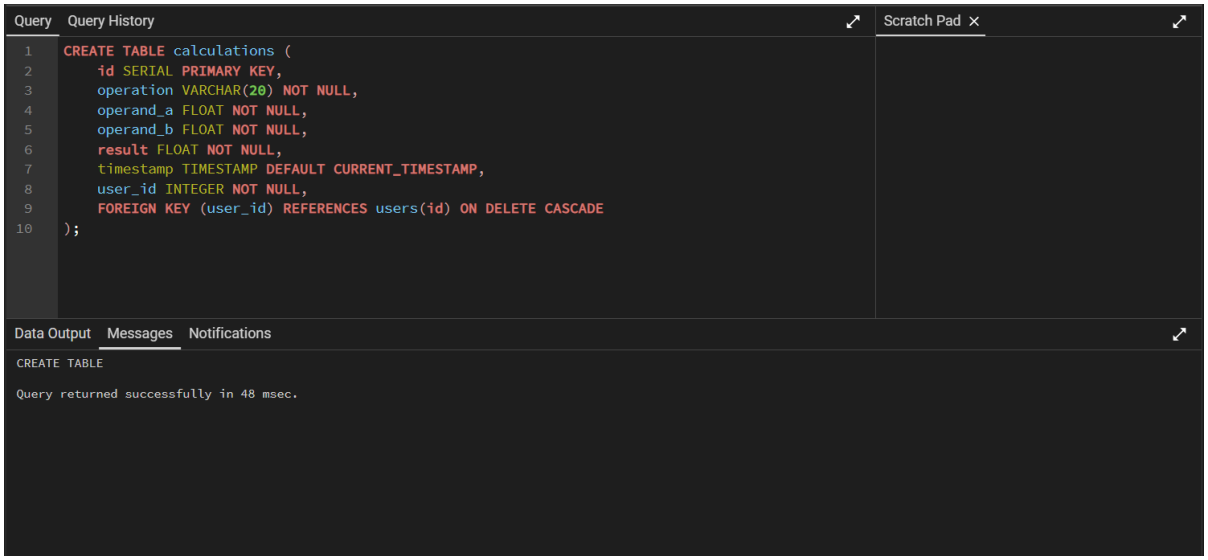


Running docker compose up. The postgres_db, fastapi_calculator and pgadmin containers all start, and postgres_db reports Healthy.

Create Tables

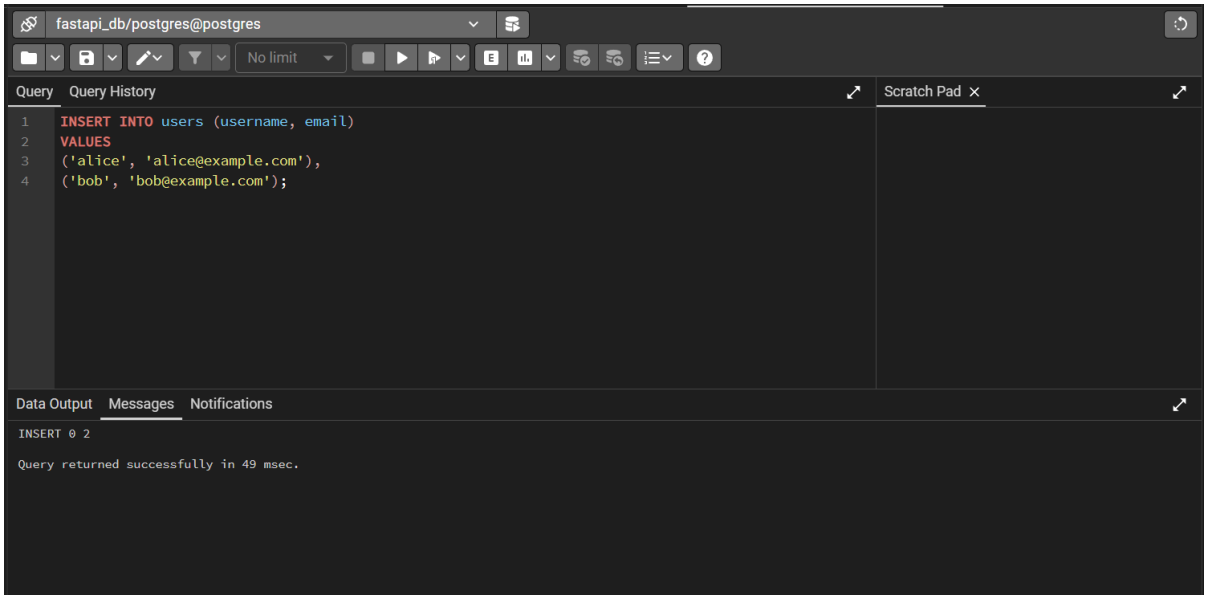


Creating the users table in the pgAdmin Query Tool. The query returned successfully.

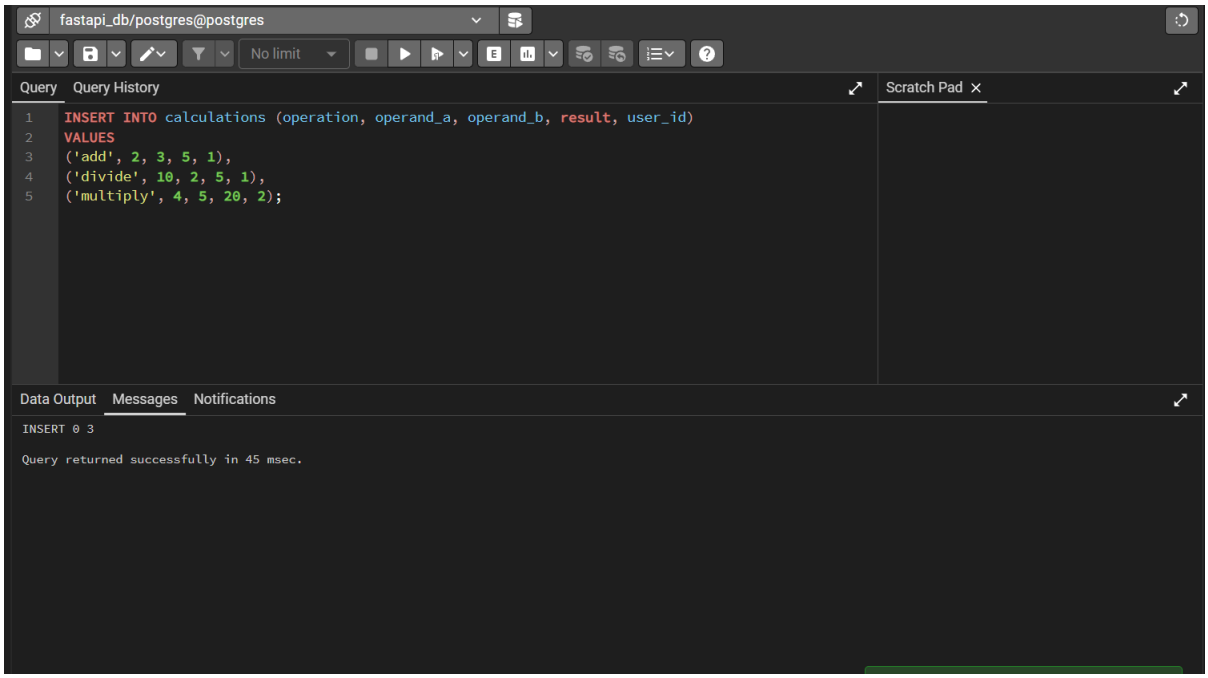


Creating the calculations table, including the foreign key from user_id to users(id) with ON DELETE CASCADE.

Insert Records

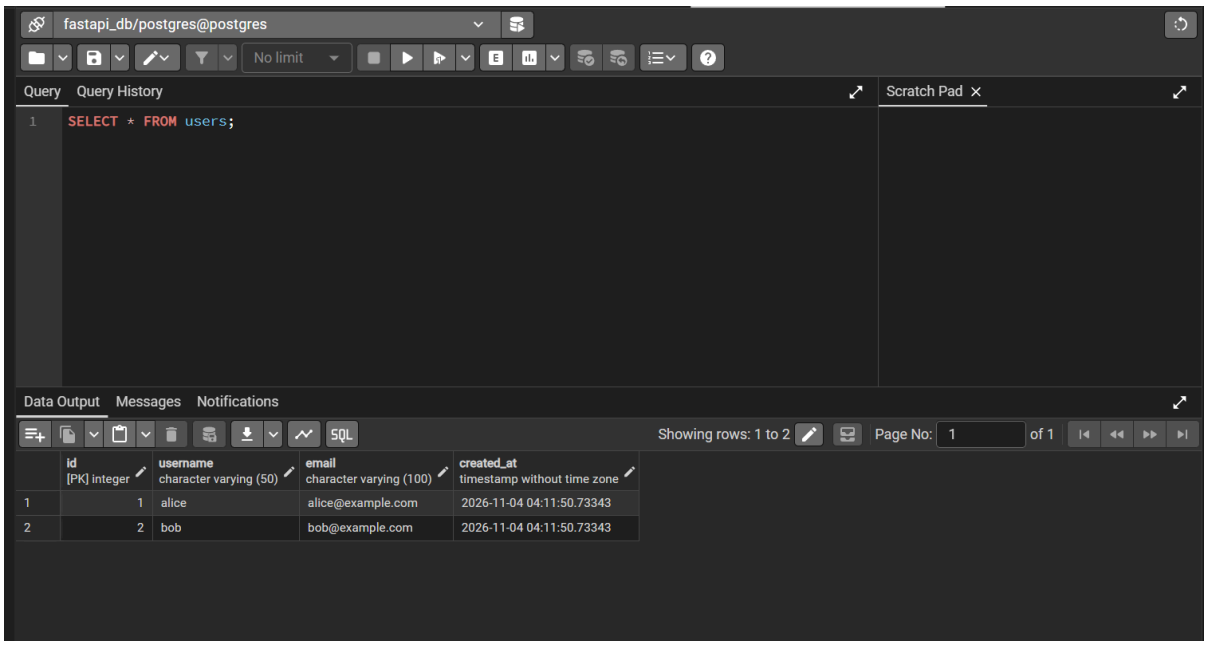


Inserting alice and bob into the users table (INSERT 0 2).



Inserting three rows into the calculations table (INSERT 0 3).

Query Data



SELECT * FROM users returns the table content: alice (id 1) and bob (id 2).

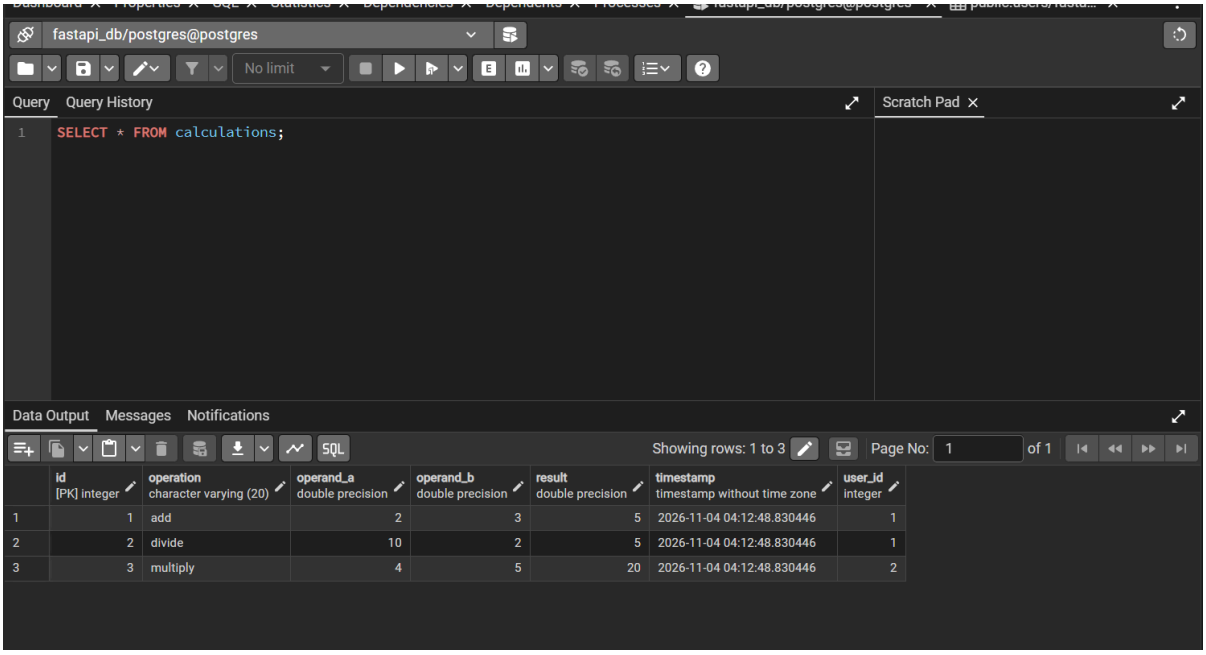
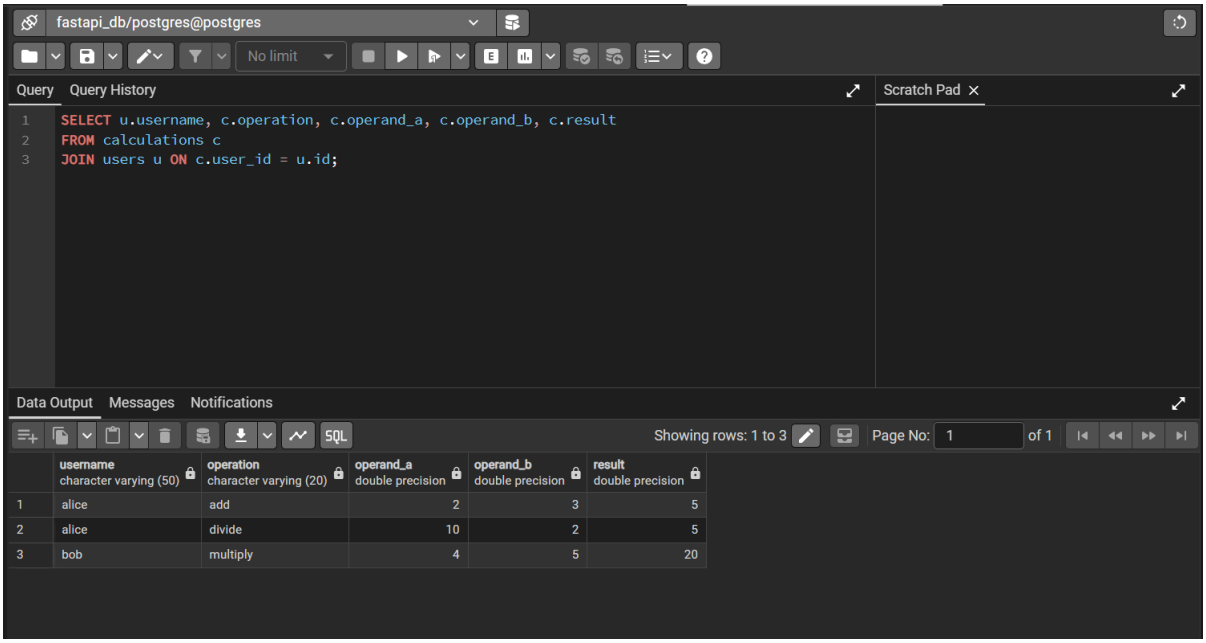
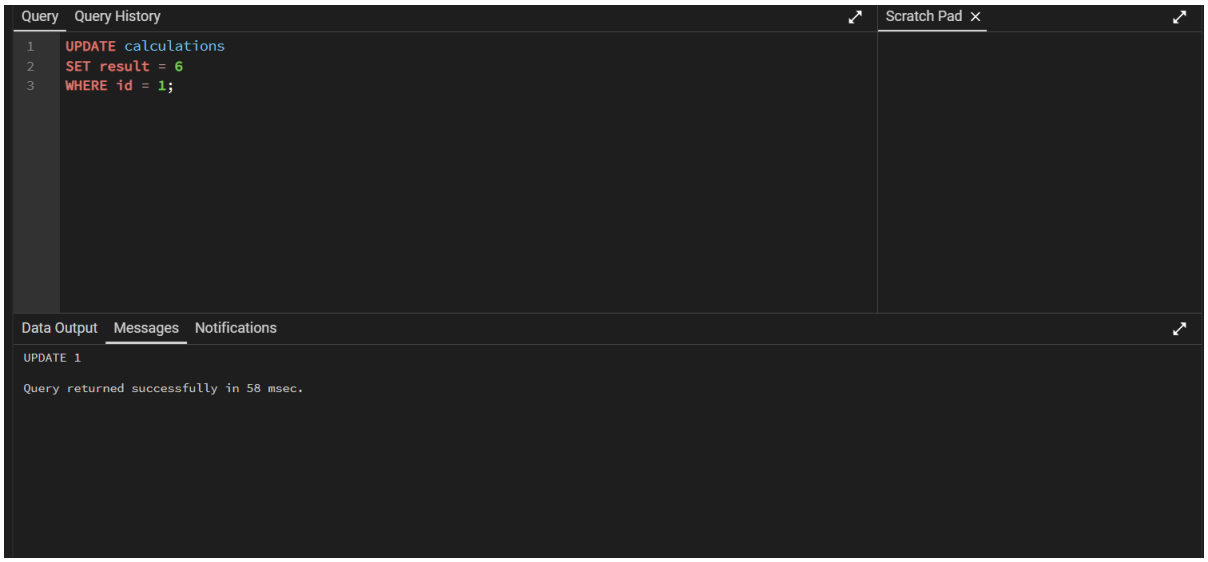


Figure 7: SELECT * FROM calculations returns all three calculation rows.



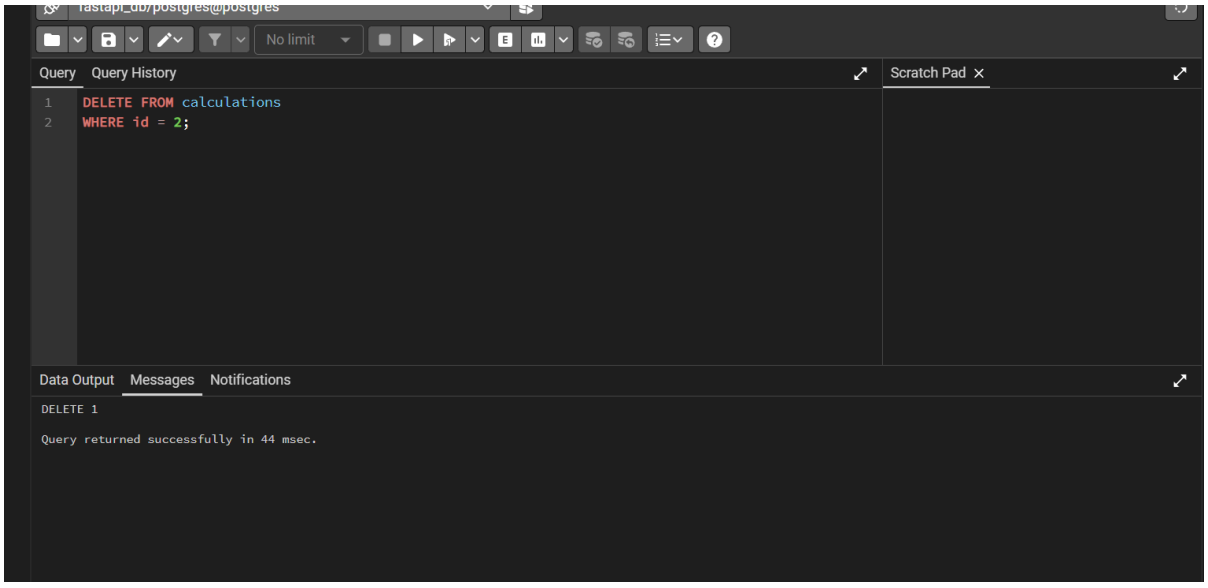
Joining users and calculations. Alice has two calculations and bob has one, which shows the one-to-many relationship.

Update a Record



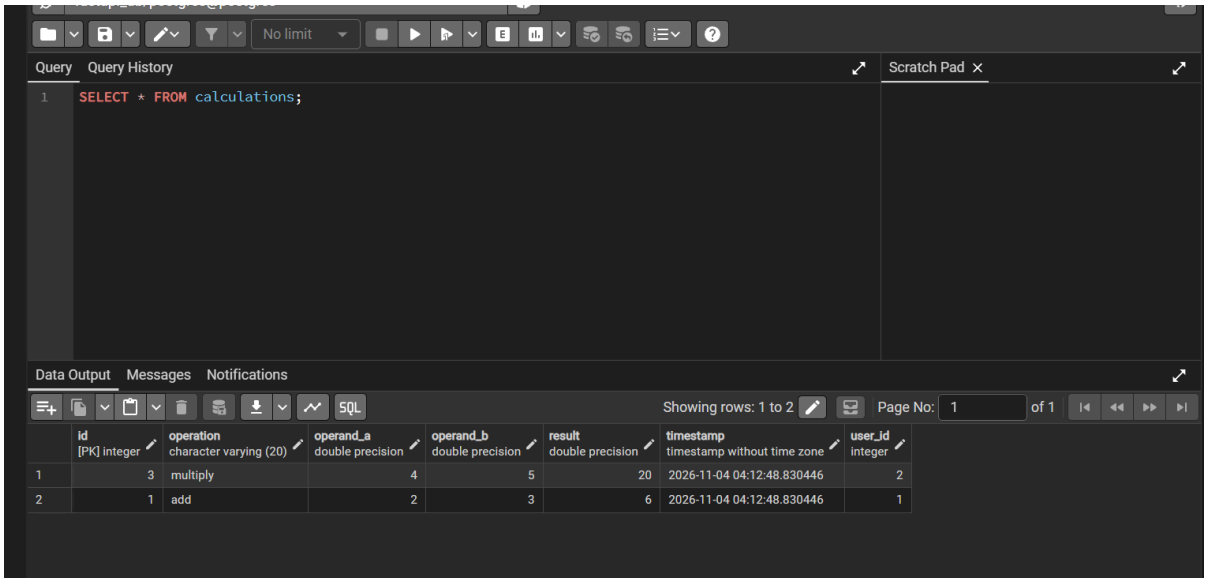
Updating the result of calculation id 1 to 6 (UPDATE 1).

Delete a Record



Deleting calculation id 2 (DELETE 1).

Final State



	id [PK] integer	operation character varying (20)	operand_a double precision	operand_b double precision	result double precision	timestamp timestamp without time zone	user_id integer
1	3	multiply	4	5	20	2026-11-04 04:12:48.830446	2
2	1	add	2	3	6	2026-11-04 04:12:48.830446	1

SELECT * FROM calculations after the update and delete. Id 1 now has result 6 and id 2 is gone.